SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications.* (BTL 5)

Assessment Tool Weightage: 50%

Duration : 1 Hour
QUESTIONS: 5 **Total Marks:50**

Annexure A

Design Task

Code of Conduct:

I Vasantha Nithi D Y (Name / Reg No) certify that this submission is my original work and that I have adhered
to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Design Task

1. **Hybrid AI Recommendation Engine Design**
   Design a hybrid AI-based recommendation system that integrates content-based and collaborative filtering techniques using PySpark RDDs. Explain how distributed data processing can be used to handle large-scale user-item interactions, and describe the role of intelligent agents in adapting recommendations based on user behavior and feedback in real time.

2. **Smart Disaster Detection System**
   Develop a distributed AI system using PySpark and RDD operations to detect natural disasters (such as floods or earthquakes) from multi-source data streams (satellite images, sensors, and social media). Describe the system architecture, data fusion techniques, and how intelligent agents coordinate to provide early warnings and response strategies.

3. **AI-Based Fraud Detection Framework**
   Design a scalable fraud detection system using PySpark RDDs to process large volumes of financial transactions. Explain how anomaly detection algorithms can be implemented in a distributed manner and how intelligent agents collaborate to identify suspicious patterns, trigger alerts, and continuously improve detection accuracy.

4. **Autonomous Supply Chain Optimization System**
   Construct a distributed AI system using PySpark for optimizing supply chain operations (inventory, logistics, demand forecasting). Describe how RDD transformations support large-scale data processing and how multiple intelligent agents interact to make decentralized decisions for cost minimization and efficiency improvement.

5. **Personalized Learning Analytics Platform**
   Design an AI-driven educational platform that uses PySpark and RDDs to analyze student learning behavior and performance data. Explain how the system supports adaptive learning through intelligent agents, real-time feedback, and scalable analytics to personalize content delivery for thousands of learners.

RUBRICS FOR CALCULATION

Evaluation Criteria	Problem Understanding & Requirement Analysis	System Architecture Design	Use of PySpark & RDD Operations	Intelligent Agent Integration	Scalability & Distributed Computing Design	Data Flow & Processing Pipeline	Innovation & Creativity	Clarity, Presentation & Documentation
Weightage	10%	20%	15%	15%	10%	10%	10%	10%

Criteria	Excellent (5)	Good (4)	Average (3)	Poor (2)
Problem Understanding & Requirement Analysis	Clearly defines problem, objectives, and constraints	Mostly clear with minor gaps	Basic understanding	Unclear or incorrect
System Architecture Design	Complete, scalable, well-structured architecture with diagrams	Good design with minor issues	Basic design, lacks detail	Poor/incomplete

Use of PySpark & RDD Operations	Efficient and correct use of RDD transformations/actions	Mostly correct usage	Limited or partial use	Incorrect/missing
Intelligent Agent Integration	Strong integration with clear agent roles and logic	Good integration	Basic integration	Poor/no integration
Scalability & Distributed Computing Design	Clearly addresses scalability, fault tolerance, parallelism	Covers most aspects	Limited consideration	Not addressed
Data Flow & Processing Pipeline	Clear, logical, and well-defined data flow	Mostly clear	Some gaps	Poorly defined
Innovation & Creativity	Highly innovative, realistic, and impactful solution	Some innovation	Basic idea	No originality
Clarity, Presentation & Documentation	Very clear, neat diagrams, well-organized report	Mostly clear	Somewhat organized	Poor presentation

CO5 – Assessment Tool 1

Smart Disaster Detection System

1. Problem Statement

Natural disasters such as floods and earthquakes require early detection and rapid response. However, disaster-related data comes from multiple sources such as **IoT sensors, satellite images, and social media**, producing large volumes of data that must be processed in real time.

The proposed system uses **PySpark, RDD operations, and intelligent agents** to process distributed data, detect potential disasters, and generate early warnings.

2. Proposed System Architecture

IoT Sensors ──────┐

Satellite Data ────┤

Social Media ─────┤

▼

Data Ingestion Layer

▼

PySpark Distributed Cluster

▼

RDD Transformations

(map → filter → flatMap)

▼

Data Fusion

▼

AI Detection Model

▼

Decision Agent

```
      /       \
      ▼       ▼
 Alert Agent   Response Agent
      \       /
       ▼     ▼
  Authorities / Users
```

3. PySpark and RDD Operations

The incoming data is processed using distributed **RDD operations**:

- map() – converts raw sensor and social-media data into a common format.
- filter() – removes invalid or irrelevant data.
- flatMap() – extracts multiple events or keywords from a data source.
- reduceByKey() – combines similar events from different sources.
- count() – counts detected disaster-related events.

Example:

```
rdd = sc.textFile("disaster_data.txt")


alerts = rdd.flatMap(lambda x: x.split()) \
    .map(lambda x: (x.lower(), 1)) \
    .reduceByKey(lambda a, b: a + b)


print(alerts.collect())
```

4. Intelligent Agent Integration

The system contains multiple intelligent agents:

Agent	Role
Sensor Agent	Collects data from sensors, satellites and social media.
Analysis Agent	Detects patterns and identifies possible disasters.
Decision Agent	Evaluates disaster severity and selects an appropriate response.
Alert Agent	Sends early warnings to authorities and affected people.
Response Agent	Coordinates emergency services and resource allocation.

5. Data Flow and Processing Pipeline

1. Data is continuously collected from multiple sources.
2. PySpark distributes the data across cluster nodes.
3. RDD transformations clean and process the data in parallel.
4. Data from multiple sources is fused to improve reliability.
5. The AI model detects disaster patterns.
6. The Decision Agent evaluates severity.
7. Alert and Response Agents generate warnings and coordinate action.

6. Scalability and Innovation

Scalability:

- PySpark distributes processing across multiple nodes.
- RDDs allow parallel processing of large datasets.
- The system can handle increasing data by adding more cluster nodes.
- Distributed architecture improves fault tolerance.

Innovation:

The system uses **multi-source data fusion and multi-agent coordination**. Instead of depending on a single data source, a disaster alert is validated using multiple signals, reducing false alarms.

7. Conclusion

The Smart Disaster Detection System combines **PySpark, RDD operations, distributed processing, AI models, and intelligent agents** to provide scalable real-time disaster detection. The system processes large volumes of multi-source data, detects potential disasters, and coordinates early warnings and emergency response.